

Giovanni Pellegrini

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I am currently working on the creation and development of cutting-edge artificial intelligence products for businesses using traditional machine learning and agentic AI systems.

Current

Mar 2023 - **AI Scientist**, *Ipazia S.p.A.*, Milan, Italy, Remote Work.

- Ongoing ○ Data preprocessing, development and training of an algorithm to identify problematic users using time-series analysis for an international gambling platform.
- Estimation of loan's recovery rate of clients for an international financial institution.
- Creation of a data preprocessing pipeline and recommendation algorithms to improve the match between candidates and job vacancies for a major Italian recruiting company.
- Research and development of internal products based on agentic AI solutions using LangGraph, Autogen.

Work Experience

Sep 2021-Mar **Machine Learning Scientist**, *GlassFORM.ai*, Rovereto, Italy.

- 2023 ○ Development and training of machine learning algorithms for automatically control the parameters of glass forming machines in real time during production.
- Presentation of AI solutions at GlassTEC 2022 fair.

Sep 2020-Feb **Teaching Assistant**, *University of Trento*, Trento.

- 2021 ○ Teaching assistant for the course *Machine Learning* at the master program of Computer Science.

Jun-Sep 2020 **Project Manager**, *TIM*, Italy, Remote work.

- Business project in collaboration with TIM on the recommendation of bundles of products and telecommunication services.

Apr-Sep 2016 **Research Intern**, *TIM*, Trento, Italy.

- Research project about the fine-grained estimation of the position of tweets.

Mar-July **Junior Developer Intern**, *Trento RISE*, Trento, Italy.

- 2014 ○ Development of a web application with AngularJS at the TechPeaks startup incubator.

Education

2017-2021 **Ph.D. student in Computer Science and Telecommunications**, *University of Trento*, Italy.

Scholarship sponsored by TIM and EIT Digital. Advisor: Andrea Passerini.
Supervision of the thesis of two Bachelor and one Master students.
Period abroad of 4 months at the University of Aalborg, Denmark.

2015-2017 **MSc. in Computer Science**, *Univeristy of Trento*, Italy, 109/110.

Thesis: *Neuro-Symbolic learning with Type Extension Trees.*

2016 **Erasmus Semester**, *Univeristy of Aalborg*, Denmark.

Host university supervisor: Manfred Jaeger.

2011-2014 **BSc. in Computer Science**, *University of Trento*, Italy.

Other Activities

- Jun 2022 **Agile - Scrum Master Course**, *Rovereto*.
40 hours training course on agile methodology with focus on scrum master activities.
- Feb 2021- **Pytorch Geometric Tutorial**, Online series.
- Mar 2022 I am one of the organizers of the Pytorch Geometric tutorial, a series of online events dedicated to the Graph Neural Network ecosystem, including both theoretical advancements and practical applications. 120,000+ visualizations on YouTube.
- Nov 2020 **Organization DA2PL workshop**, *Italy*, Online conference.
Organization committee of the DA2PL 'From Multiple Criteria Decision Aid to Preference Learning' workshop.
- Feb 2020 **Synth Research Camp**, *Germany*.
Winter school on the latest research advancements on constraint learning, statistical relational learning and meta learning.
- Jun 2018 **Summer School on Artificial Intelligence and Games**, *Greece*.
First summer school on research and business applications of machine learning and artificial intelligence algorithms for game content generation, agent learning and user behaviour understanding.

Latest Publications

- 2021 **Learning Aggregation Functions.**
Pellegrini G., Tibo A., Frasconi P., Passerini A., Jaeger M.
Accepted at IJCAI '21.
- 2019 **Counts-of-counts similarity for prediction and search in relational data.**
Jaeger M., Lippi M., **Pellegrini G.**, Passerini A.
Data mining and knowledge discovery Journal.
- 2018 **No more ready-made deals: constructive recommendation for telco service bundling.**
Dragone P., **Pellegrini G.**, Vescovi M., Tentori K., Passerini A.
Proceedings of the 12th ACM Conference on Recommender Systems.

Certificates

- 2014 **edX Honor Code Certificate for Learning From Data (Introductory Machine Learning)**, *edX*.
Introductory course on machine learning on the online platform edX.

Hard Skills

- AI GRAPH NEURAL NETWORKS, GENERATIVE AI, AGENTIC AI, LARGE LANGUAGE MODELS, PROMPT ENGINEERING, RAG, STATISTICAL RELATIONAL LEARNING, PREFERENCE LEARNING
- Systems LINUX, MACOS, WINDOWS
- Languages PYTHON, JAVA, SQL
- Tools PYTORCH, PYTORCH GEOMETRIC, NUMPY, GIT, LANGGRAPH, AUTOGEN, JIRA, AWS, MICROSOFT OFFICE SUITE, SCIKIT-LEARN, PANDAS, POLARS, MATPLOTLIB, TENSORFLOW

Soft Skills

TEAMWORK, PROBLEM SOLVING, ANALYTICAL THINKING, TEACHING AND PRESENTATION, INTERACTION WITH CLIENTS, ADAPTABILITY, CREATIVITY

Languages

Italian Native
English Professional-Academic